Lecture 15 (*March 24*)

Reading: Section 4.4
Viewing: Lecture 17

We finished the previous Lecture with an example that basically asked you to "pick an orthonormal basis" of a given vector space. We'll now see how to explicitly construct such a basis, and even more so, we'll give an explicit algorithm to construct an orthonormal basis from any given basis. This is called the **Gram-Schmidt process** and here's how it goes: you start from a general collection of linearly independent vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ (in whatever vector space you are working with) and the goal is to "make" them orthonormal. You are allowed to modify the individual vectors $\boldsymbol{v}_i$ by rescaling them and adding linear combinations of the other vectors:

$$\boldsymbol{v}_i \quad \rightsquigarrow \quad \alpha_1 \boldsymbol{v}_1 + \cdots + \alpha_i \boldsymbol{v}_i + \cdots + \alpha_n \boldsymbol{v}_n \quad \text{with } \alpha_i \neq 0$$

since these operations do not change the vector space spanned by $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$. Start by renormalizing $\boldsymbol{v}_1$ so that it has length 1:

$$\boxed{\text{Step 1 :}} \quad \boldsymbol{v}_1 \rightsquigarrow \boldsymbol{q}_1 = \frac{\boldsymbol{v}_1}{||\boldsymbol{v}_1||}$$

Then modify $\boldsymbol{v}_2$ so that it is perpendicular to $\boldsymbol{q}_1$, and then renormalize it to have length 1:

$$\boxed{\text{Step 2 :}} \quad \boldsymbol{v}_2 \rightsquigarrow \boldsymbol{w}_2 = \boldsymbol{v}_2 - \text{proj}_{\boldsymbol{q}_1} \boldsymbol{v}_2 \quad \text{and} \quad \boldsymbol{w}_2 \rightsquigarrow \boldsymbol{q}_2 = \frac{\boldsymbol{w}_2}{||\boldsymbol{w}_2||}$$

Then modify $\boldsymbol{v}_3$ so that it is perpendicular to $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$, and then renormalize it to have length 1:

$$\boxed{\text{Step 3 :}} \quad \boldsymbol{v}_3 \rightsquigarrow \boldsymbol{w}_3 = \boldsymbol{v}_3 - \text{proj}_{\boldsymbol{q}_1} \boldsymbol{v}_3 - \text{proj}_{\boldsymbol{q}_2} \boldsymbol{v}_3 \quad \text{and} \quad \boldsymbol{w}_3 \rightsquigarrow \boldsymbol{q}_3 = \frac{\boldsymbol{w}_3}{||\boldsymbol{w}_3||}$$

... so far and so forth, until the last step where we modify $\boldsymbol{v}_n$ so that it is perpendicular to $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, $\ldots$, $\boldsymbol{q}_{n-1}$ and then renormalize it to have length 1:

$$\boxed{\text{Step } n \text{ :}} \quad \boldsymbol{v}_n \rightsquigarrow \boldsymbol{w}_n = \boldsymbol{v}_n - \text{proj}_{\boldsymbol{q}_1} \boldsymbol{v}_n - \cdots - \text{proj}_{\boldsymbol{q}_{n-1}} \boldsymbol{v}_n \quad \text{and} \quad \boldsymbol{w}_n \rightsquigarrow \boldsymbol{q}_n = \frac{\boldsymbol{w}_n}{||\boldsymbol{w}_n||}$$

At the end of this procedure, all the vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ are orthogonal to each other by construction, and they all have length 1. This is because at each step, we subtract from every $\boldsymbol{v}_i$ its projection onto the already constructed vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{i-1}$. What remains will be orthogonal to these vectors. The various projections that one must calculate in this process, namely $\text{proj}_{\boldsymbol{q}_j} \boldsymbol{v}_i$ can be computed using formula (95) (which is even simpler in our case, since the denominator of (95) is equal to 1).

Example 4. *Let us construct an orthonormal basis of the vector space V of* (125)*. First pick any basis, i.e. any two vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in the vector space V (being in the vector space V just means that the coordinates of these vectors should sum up to 0) which are not multiples of each other, say:*

$$\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \quad \textit{and} \quad \boldsymbol{v}_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

Step 1 is to rescale $\boldsymbol{v}_1$ in order for it to have length 1; the way to do so is to divide $\boldsymbol{v}_1$ by its length, which is $||\boldsymbol{v}_1|| = \sqrt{14}$:

$$\boldsymbol{q}_1 = \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

Then we need to modify $\boldsymbol{v}_2$ in such a way as to be perpendicular to $\boldsymbol{q}_1$. As we have seen in Step 2 above, the way to do so is to subtract from $\boldsymbol{v}_2$ its projection onto $\boldsymbol{q}_1$:

$$\boldsymbol{w}_2 = \boldsymbol{v}_2 - \mathrm{proj}_{\boldsymbol{q}_1}\boldsymbol{v}_2 \overset{(95)}{=} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} - \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \frac{1}{\sqrt{14}} \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

(the reason why the denominator of (95) doesn't show up in the formula above is that $||\boldsymbol{q}_1|| = 1$ by construction). Explicit computation shows that the vector above is:

$$\boldsymbol{w}_2 = \frac{1}{14} \begin{bmatrix} -5 \\ 4 \\ 1 \end{bmatrix}$$

The last thing we need to do is to rescale $\boldsymbol{w}_2$ in order for it to have length 1, which we do by dividing it by its length $||\boldsymbol{w}_2|| = \sqrt{3/14}$:

$$\boldsymbol{q}_2 = \frac{1}{\sqrt{42}} \begin{bmatrix} -5 \\ 4 \\ 1 \end{bmatrix}$$

The vectors $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$ computed above determine an orthonormal basis of the vector space V: it is easy to see that they both lie in V, that they both have length 1, and that they are perpendicular.

The example above shows two important things about the Gram-Schmidt process. Firstly, it can produce some square roots, so don't be scared of getting your hands a bit dirty with the computation, because the final answer might turn out to be reasonably nice. And secondly, there exist many possible choices of orthonormal bases of a vector space, e.g. just compare the vectors we guessed in (126) with those produced by the example above.

Let's see what the Gram-Schmidt process tells us about the relation between the matrix:

$$A = \left[\ \boldsymbol{v}_1 \mid \dots \mid \boldsymbol{v}_n\ \right] \quad \text{and} \quad Q = \left[\ \boldsymbol{q}_1 \mid \dots \mid \boldsymbol{q}_n\ \right]$$

where $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ are the input vectors and $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ are the output vectors of Gram-Schmidt. At each step, we are modifying the i–th column of a matrix by subtracting from it a linear combination of the previous columns, and then multiplying it by a scalar. This is doing the same thing for columns as row operations were doing for rows. Therefore, it shouldn't surprise you that:

- Adding the j–th column times λ to the i–th column of A is achieved by multiplying the latter on the right with an elimination matrix:

$$A \rightsquigarrow AE_{ji}^{(\lambda)} \qquad \text{where} \qquad E_{ji}^{(\lambda)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \lambda & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

  The entry λ is on the j–th row and i–th column (in the formula above, $i = 4$ and $j = 2$).

- Multiplying the i–th column of A by the scalar λ is achieved by multiplying A on the right with a diagonal matrix:

$$A \rightsquigarrow AD_i^{(\lambda)} \qquad \text{where} \qquad D_i^{(\lambda)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

  The entry λ is on i–th row (in the formula above, $i = 2$).

Compare these with the operations you learned in Lecture 2 when discussing Gaussian elimination and LU factorizations: the upshot is that while row operations correspond to multiplication on the left, column operations correspond to multiplication on the right. Therefore, we conclude that the connection between the input and the output of Gram-Schmidt is:

$$Q = A \underbrace{D_1^{(\mu_1)}}_{\text{first step}} \underbrace{E_{12}^{(\lambda_{12})} D_2^{(\mu_2)}}_{\text{second step}} \underbrace{E_{13}^{(\lambda_{13})} E_{23}^{(\lambda_{23})} D_3^{(\mu_3)}}_{\text{third step}} \dots \underbrace{E_{1n}^{(\lambda_{1n})} \dots E_{n-1,n}^{(\lambda_{n-1,n})} D_n^{(\mu_n)}}_{\text{last step}} \tag{127}$$

where the scalars μ_i and λ_{ij} are determined by the orthonomality requirement at every stage of Gram-Schmidt. By moving all the E's and D's in the other side of the equation, we obtain:

$$A = QD_n^{\left(\frac{1}{\mu_n}\right)} E_{n-1,n}^{(-\lambda_{n-1,n})} \dots E_{1n}^{(-\lambda_{1n})} \dots D_3^{\left(\frac{1}{\mu_3}\right)} E_{23}^{(-\lambda_{23})} E_{13}^{(-\lambda_{13})} D_2^{\left(\frac{1}{\mu_2}\right)} E_{12}^{(-\lambda_{12})} D_1^{\left(\frac{1}{\mu_1}\right)}$$

If we multiply the D and E matrices together, we will obtain an upper triangular matrix R. Hence:

Fact 10. *Any matrix A with linearly independent columns can be uniquely written as:*

$$\boxed{A = QR} \tag{128}$$

where the columns of Q are orthonormal and R is an upper triangular square matrix:

$$R = \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}$$

Note that this result also makes sense for non-square matrices A. As we have seen in (129)*, computing the QR factorization of a matrix is achieved by performing Gram-Schmidt on its columns.*

MIDTERM 1 *(March 26)*